Delay propagation on a suburban railway network

Guillaume Dalle[1], Yohann De Castro[2], Axel Parmentier[1]

[1] CERMICS - École des Ponts ParisTech
[2] Institut Camille Jordan - École Centrale de Lyon

ROADEF 2020 - Montpellier

Table of Contents

1. Introduction
2. Model and inference
3. Statistical analysis
4. Numerical tests

Context of the study

Context of the study

Context of the study

- Partnership with French railway company SNCF

Context of the study

- Partnership with French railway company SNCF
- Goal: create a **delay prediction model** for the Paris suburban network

Context of the study

- Partnership with French railway company SNCF
- Goal: create a **delay prediction model** for the Paris suburban network
- Applications:
  - information
  - regulation
  - scheduling

Context of the study

- Partnership with French railway company SNCF
- Goal: create a **delay prediction model** for the Paris suburban network
- Applications:
  - information
  - regulation
  - scheduling
- Problem: we **cannot consider trains individually**

What is delay propagation?

- Trains share **resources**: infrastructure, driver, rolling stock, etc.

What is delay propagation?

- Trains share **resources**: infrastructure, driver, rolling stock, etc.
- Primary delay + resource conflit = propagated delay

What is delay propagation?

- Trains share **resources**: infrastructure, driver, rolling stock, etc.
- Primary delay + resource conflit = propagated delay

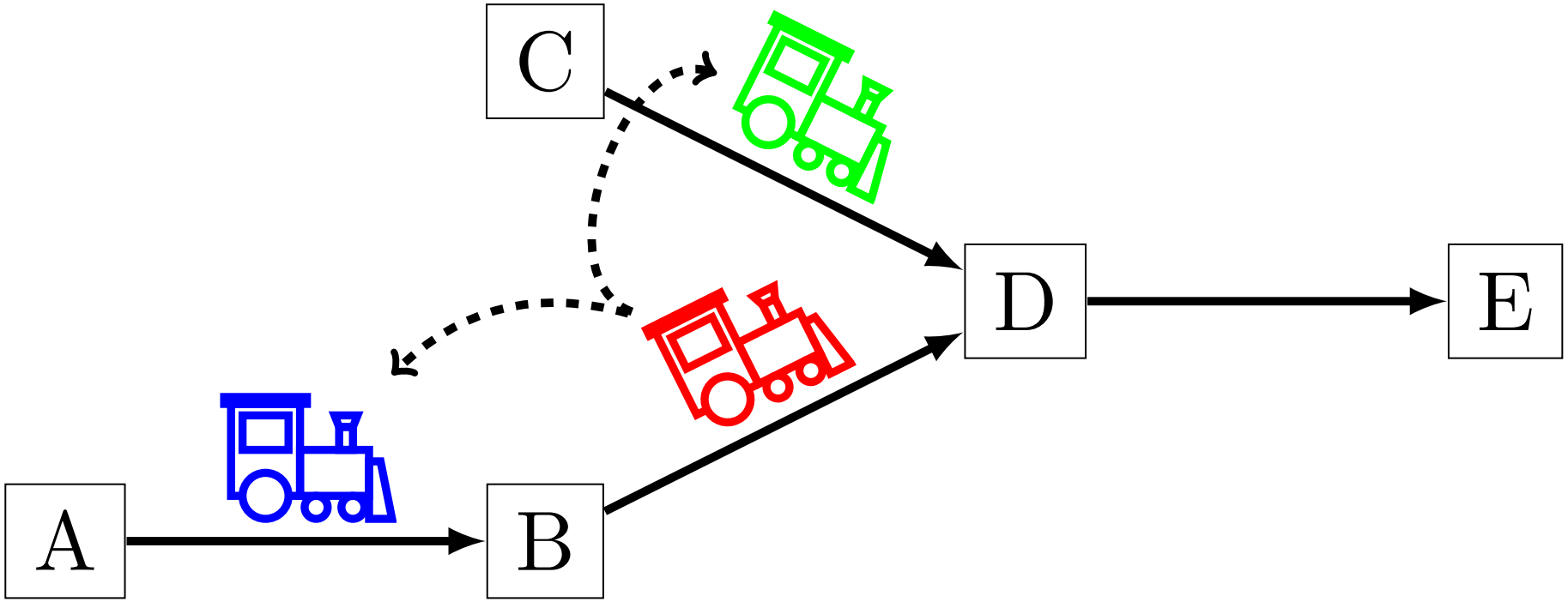


Figure 1: Delay propagation caused by infrastructure conflict

Figure inspired by Monechi et al. (2018)

Available data and consequences

- 2 years of event times (arrivals & departures) for RER & Transilien

Available data and consequences

- 2 years of event times (arrivals & departures) for RER & Transilien
- Standard methods use explicit dependencies between events (see Corman and Kecman, 2018).

Available data and consequences

- 2 years of event times (arrivals & departures) for RER & Transilien

- Standard methods use explicit dependencies between events (see Corman and Kecman, 2018). Inadequate here because:
  - **Unknown conflicts** for resources
  - **Unpredictable modifications** in real time

Available data and consequences

- 2 years of event times (arrivals & departures) for RER & Transilien
- Standard methods use explicit dependencies between events (see Corman and Kecman, 2018). Inadequate here because:
  - **Unknown conflicts** for resources
  - **Unpredictable modifications** in real time
- Solution: replace these dependencies with **latent interactions carried by the infrastructure**

Delay decomposition

- Journey of train n: path through a graph.

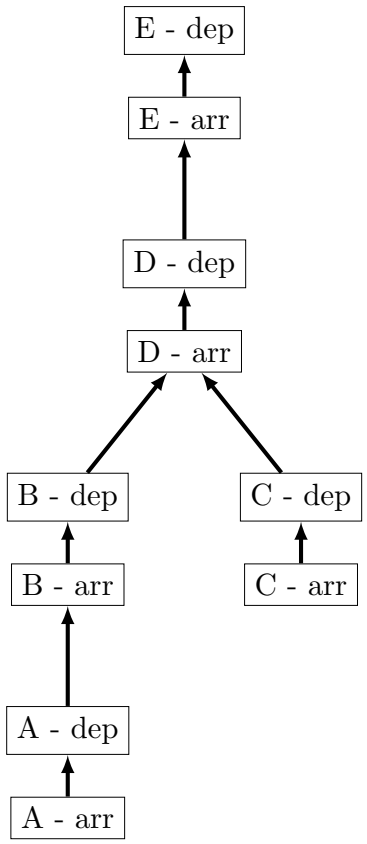


Figure 2: Milestones graph $G = (V, E)$ linking events

Delay decomposition

- Journey of train n: path through a graph.
  Oservation $H^d_{n,v}$: time of event v on day d

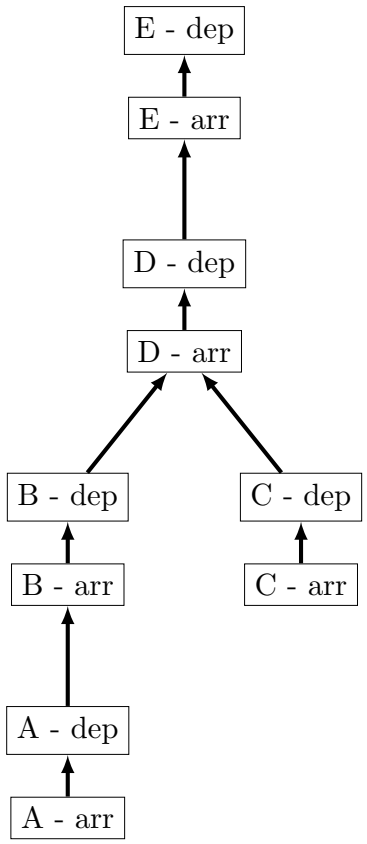


Figure 2: Milestones graph $G = (V, E)$ linking events

Delay decomposition

- Journey of train n: path through a graph.
  Oservation $H^d_{n,v}$: time of event v on day d
- When crossing edge e at time t, every train suffers:
  - A collective delay $J^d_{t,e}$, the **network jam**
  - An individual delay $I^d_{n,e}$

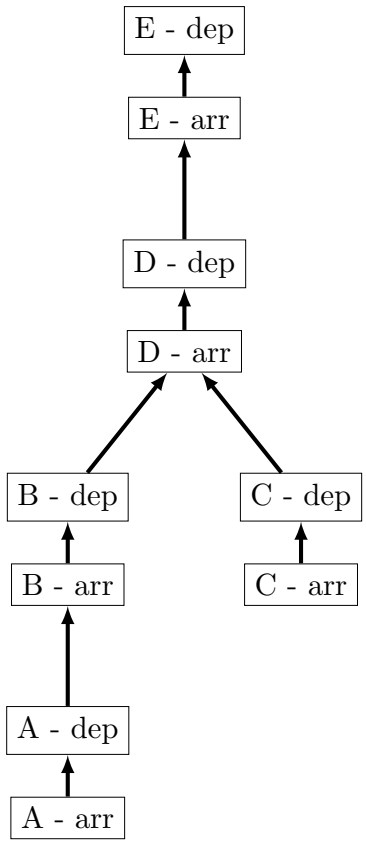


Figure 2: Milestones graph $G = (V, E)$ linking events

Delay decomposition

- Journey of train n: path through a graph.
  Oservation $H^d_{n,v}$: time of event v on day d
- When crossing edge e at time t, every train suffers:
  - A collective delay $J^d_{t,e}$, the **network jam**
  - An individual delay $I^d_{n,e}$

$$\underbrace{H^d_{n,v+1} - H^d_{n,v}}_{\text{realized duration of edge } e} = \underbrace{h^d_{n,v+1} - h^d_{n,v}}_{\text{scheduled duration of edge } e} + \underbrace{J^d_{t,e}}_{\text{network jam on edge } e \text{ at time } t} + \underbrace{I^d_{n,e}}_{\text{individual delay}}$$

with $t = \left\lfloor H^d_{n,v} \right\rfloor$

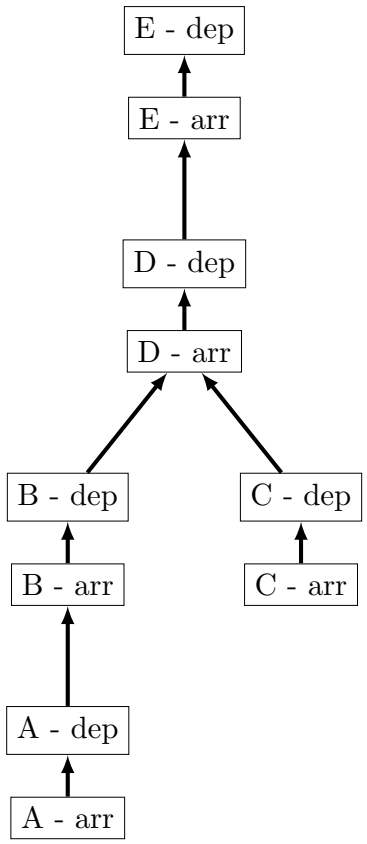


Figure 2: Milestones graph $G = (V, E)$ linking events

Network jam propagation

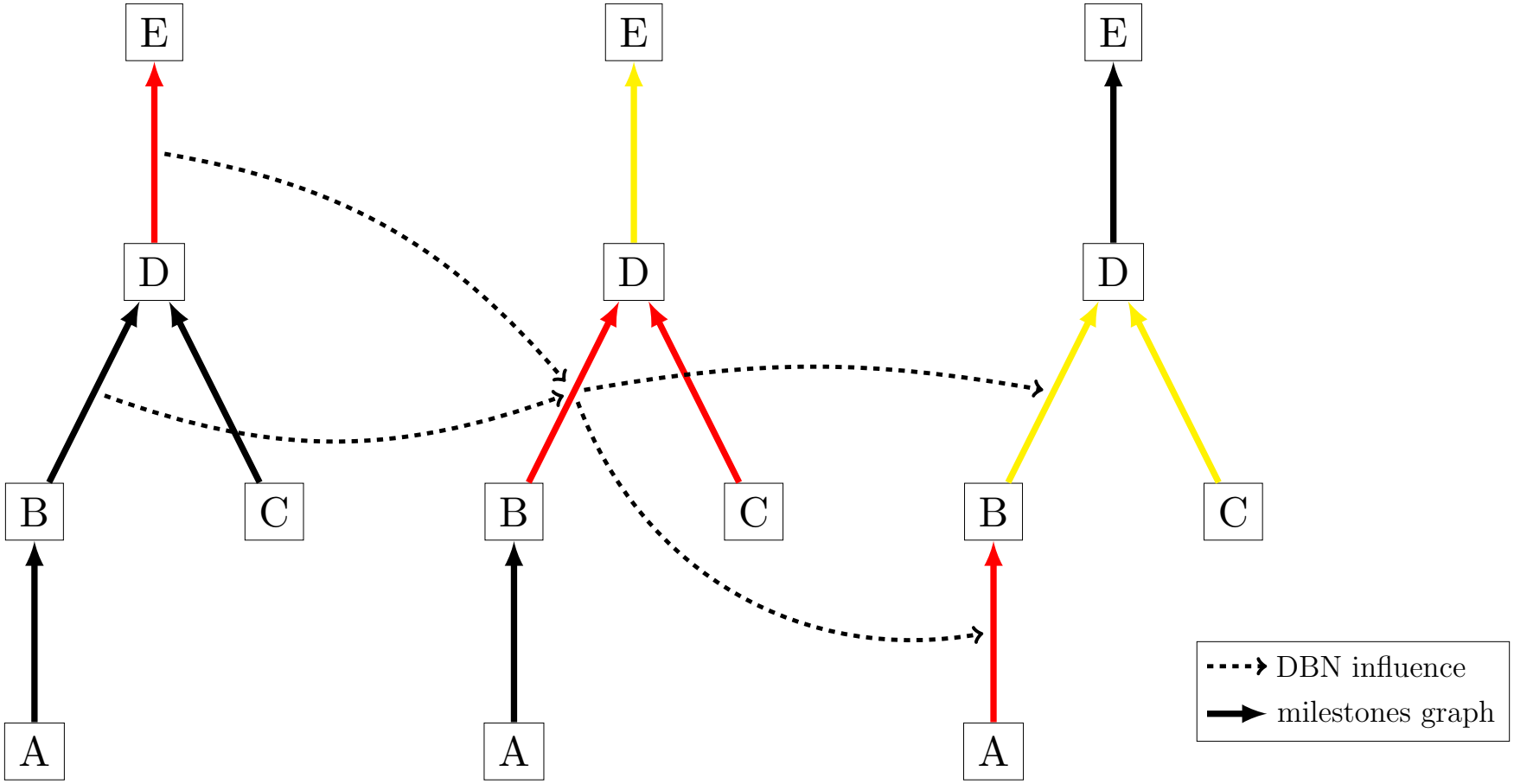
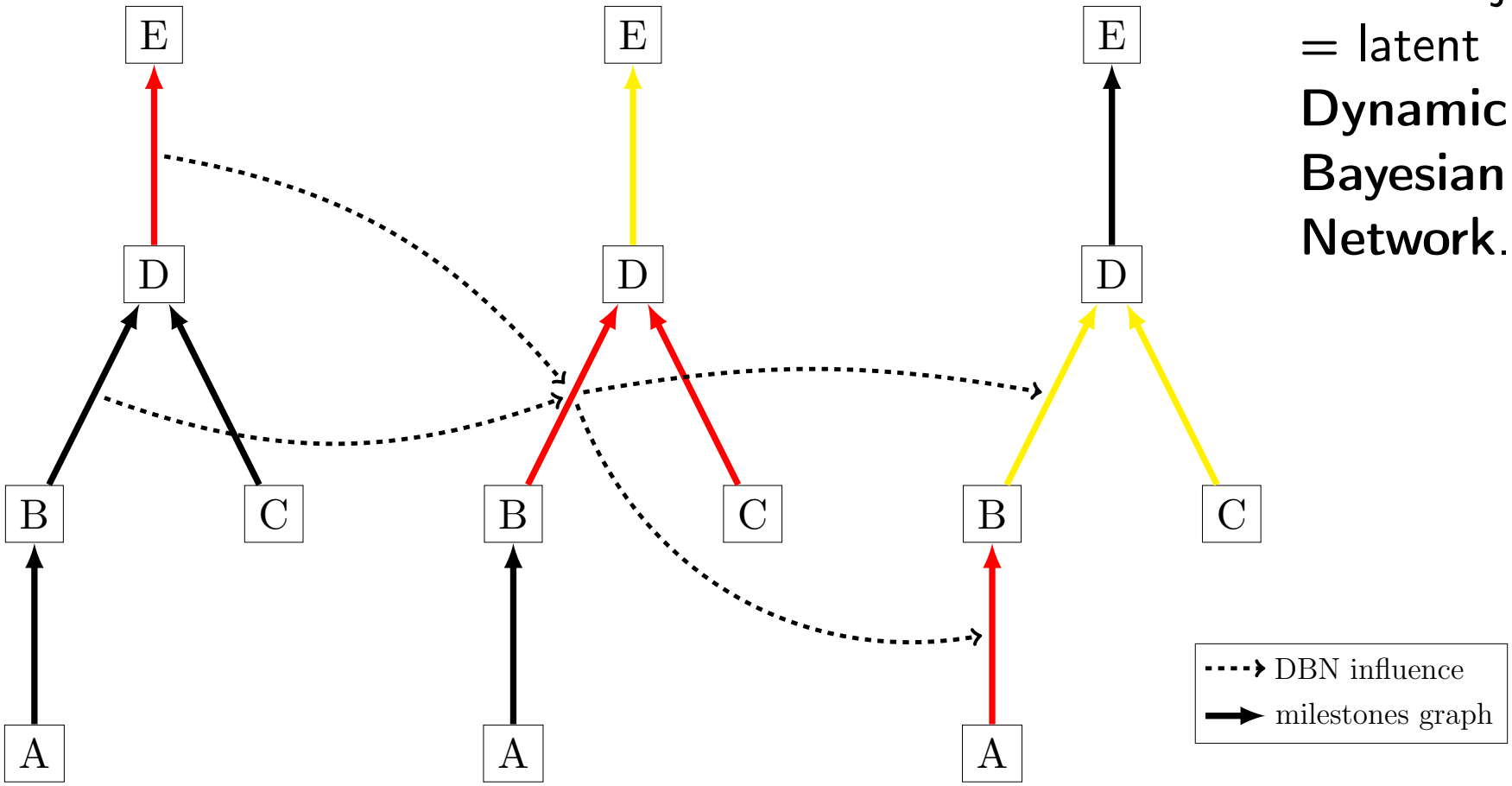


Figure 3: Propagation of the network jam

Network jam propagation



Network jam = latent **Dynamic Bayesian Network**.

Figure 3: Propagation of the network jam

A flexible framework

- Nearly everything can (and should) be **tuned**:

A flexible framework

- Nearly everything can (and should) be **tuned**:
  - For the network jam:
    - Dependency structure
    - Conditional distributions

A flexible framework

- Nearly everything can (and should) be **tuned**:
  - For the network jam:
    - Dependency structure
    - Conditional distributions
  - For the individual delay:
    - Timetable corrections
    - Exogenous influences

A flexible framework

- Nearly everything can (and should) be **tuned**:
  - For the network jam:
    - Dependency structure
    - Conditional distributions
  - For the individual delay:
    - Timetable corrections
    - Exogenous influences
- Key idea: **separate model specification from inference procedure**

A flexible framework

- Nearly everything can (and should) be **tuned**:
  - For the network jam:
    - Dependency structure
    - Conditional distributions
  - For the individual delay:
    - Timetable corrections
    - Exogenous influences
- Key idea: **separate model specification from inference procedure**
  - Stochastic variational inference (Blei et al., 2017)
  - Implementation using Pyro (Bingham et al., 2019)

A statistical question

- Can we retrieve the **network jam evolution parameters** with our amount of data?

A statistical question

- Can we retrieve the **network jam evolution parameters** with our amount of data?

- Problem: the variational inference routine is a black box

A statistical question

- Can we retrieve the **network jam evolution parameters** with our amount of data?
- Problem: the variational inference routine is a black box
- Workaround: bound the minimax risk, i.e. the **worst-case error of the best estimator**

$$\mathfrak{M} = \inf_{\hat{\theta}} \sup_{\theta^*} \mathbb{E}_\theta \left[\|\hat{\theta}(X) - \theta^*\|^2 \right]$$

as a function of model parameters and data size.

Simplifying the model

- Choose a simple network jam dynamic (linear gaussian):

$$J_{t+1}^d = W J_t^d + \mathcal{N}_E(0, \sigma^2)$$

Simplifying the model

- Choose a simple network jam dynamic (linear gaussian):

$$J^d_{t+1} = W J^d_t + \mathcal{N}_E(0, \sigma^2)$$

- Train journeys are a complicated observation mechanism!

$$\underbrace{(H^d_{n,v+1} - H^d_{n,v}) - (h^d_{n,v+1} - h^d_{n,v})}_{\Delta^d_{n,e}} = J^d_{\lfloor H^d_{n,v} \rfloor, e} + I^d_{n,e}$$

Simplifying the model

- Choose a simple network jam dynamic (linear gaussian):

$$J^d_{t+1} = W J^d_t + \mathcal{N}_E(0, \sigma^2)$$

- Train journeys are a complicated observation mechanism!

$$\underbrace{(H^d_{n,v+1} - H^d_{n,v}) - (h^d_{n,v+1} - h^d_{n,v})}_{\Delta^d_{n,e}} = J^d_{\lfloor H^d_{n,v} \rfloor, e} + I^d_{n,e}$$

- Replace them with a random subset $\mathscr{A}$ of independent noisy samples:

$$\Delta^d = \mathrm{mask}_{\mathscr{A}}(J^d) + I^d \qquad I^d \overset{\text{iid}}{\sim} \mathcal{N}_A(0, \omega^2)$$

Theorem and proof sketch

Theorem: Minimax lower bound

$$\mathfrak{M} \geq \min\left\{c_1, c_2\left(1+\frac{\omega^2}{\sigma^2}\right)^2, c_3\left(1+\frac{\omega^2}{\sigma^2}\right)\frac{E^3}{DO}\right\}$$

- D: nb of days
- E: nb of edges
- σ: signal std
- ω: noise std
- O: nb of observations

Theorem and proof sketch

Theorem: Minimax lower bound

$$\mathfrak{M} \geq \min \left\{ c_1, c_2 \left(1 + \frac{\omega^2}{\sigma^2}\right)^2, c_3 \left(1 + \frac{\omega^2}{\sigma^2}\right) \frac{E^3}{DO} \right\}$$

- D: nb of days
- E: nb of edges
- σ: signal std
- ω: noise std
- O: nb of observations

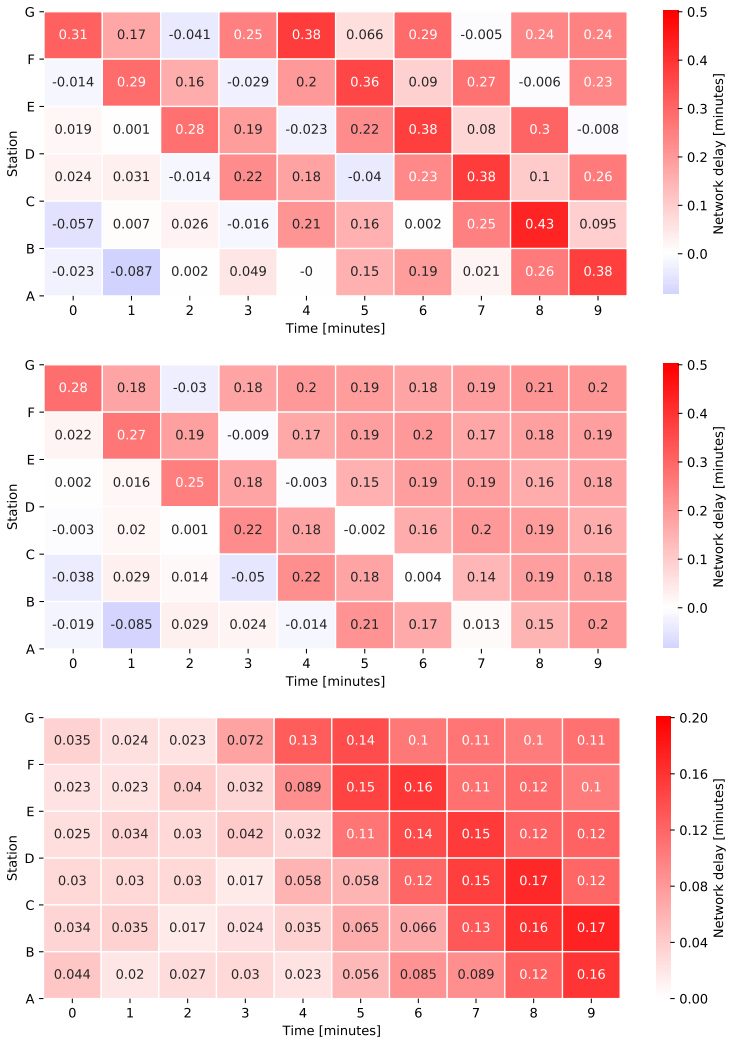
Proof idea: **similar parameters generate similar distributions**.

1. Use tedious linear algebra to upper-bound

$$\mathrm{KL}(\mathbb{P}_{W_0} \mid\mid \mathbb{P}_{W_1}) \leq f(\|W_0 - W_1\|_F)$$

2. Use Fano's method (Wainwright, 2019) to lower-bound the minimax risk using information theory

Simulated dataset: no problem



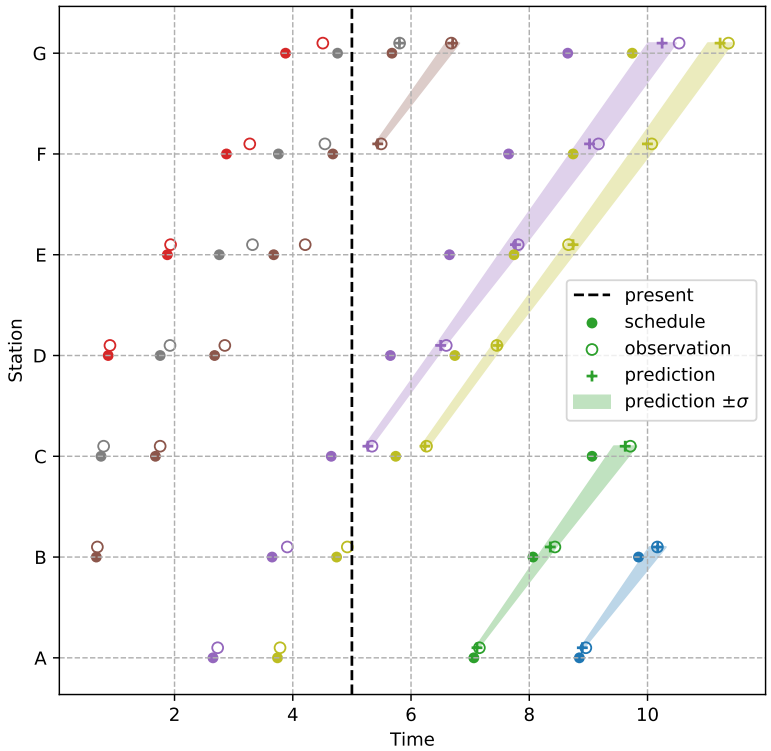


Figure 4:
Left: Inference for network jam
Above: Inference for event times

Real dataset: still disappointing

- Large computation times: need to reduce data size
  - select network subregion
  - focus on peak hour

Real dataset: still disappointing

- Large computation times: need to reduce data size
  - select network subregion
  - focus on peak hour
- Training set: 2/3 of a year / Test set: 1/3 of a year
  Observe 7:30-8:30 and predict 8:30-9:30

Real dataset: still disappointing

- Large computation times: need to reduce data size
  - select network subregion
  - focus on peak hour
- Training set: 2/3 of a year / Test set: 1/3 of a year
  Observe 7:30-8:30 and predict 8:30-9:30
- Results:
  - Individual delay improves prediction by reducing bias
  - Network jam does not seem to bring much

Real dataset: still disappointing

- Large computation times: need to reduce data size
  - select network subregion
  - focus on peak hour
- Training set: 2/3 of a year / Test set: 1/3 of a year
  Observe 7:30-8:30 and predict 8:30-9:30
- Results:
  - Individual delay improves prediction by reducing bias
  - Network jam does not seem to bring much
- Why? Limits of Pyro, slow computation, naive restricted model...

Perspectives

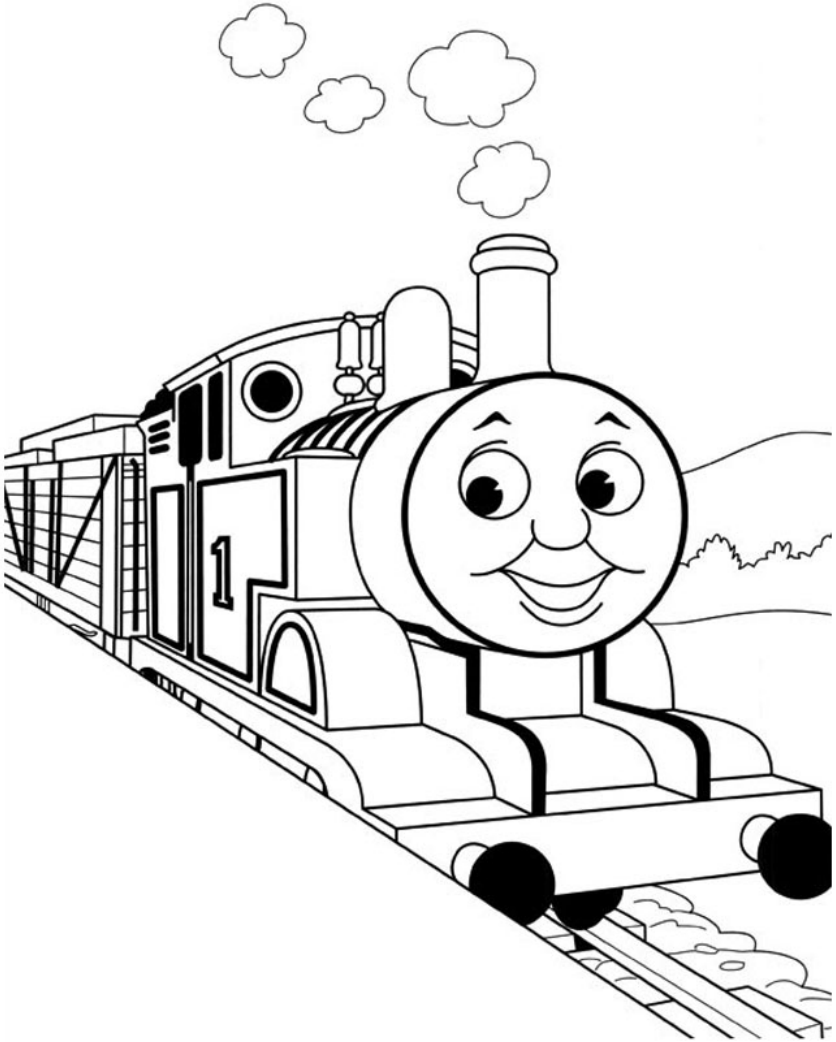
- Improving the **model**:
  - More features
  - Continuous-time version?

Perspectives

- Improving the **model**:
  - More features
  - Continuous-time version?
- Improving the **algorithm**:
  - Decompose and recompose
  - Choice of distributions

Perspectives

- Improving the **model**:
  - More features
  - Continuous-time version?
- Improving the **algorithm**:
  - Decompose and recompose
  - Choice of distributions
- More extensive tests, if possible on **open data**



Thank you for your attention

References

Bingham, E., Chen, J. P., Jankowiak, M., Obermeyer, F., Pradhan, N., Karaletsos, T., Singh, R., Szerlip, P., Horsfall, P., and Goodman, N. D. (2019). Pyro: Deep Universal Probabilistic Programming. *Journal of Machine Learning Research*, 20(28):1–6.

Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. (2017). Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association*, 112(518):859–877.

Corman, F. and Kecman, P. (2018). Stochastic prediction of train delays in real-time using Bayesian networks. *Transportation Research Part C: Emerging Technologies*, 95:599–615.

Monechi, B., Gravino, P., Di Clemente, R., and Servedio, V. D. P. (2018). Complex delay dynamics on railway networks from universal laws to realistic modelling. *EPJ Data Science*, 7(1):35.

Wainwright, M. J. (2019). *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press. Google-Books-ID: IluHDwAAQBAJ.

Train image: https://www.coloriages-pour-enfants.net/coloriages-personnages-tv/coloriage-thomas-ses-amis/?image=thomas-et-ses-amis__coloriage-thomas-et-ses-amis-train-15__1